

# F-TALENT CODE 2026

## Frequency Assignment

Time limit: 1.00 s | Memory limit: 512 MB

### Problem Statement

A telecommunications company is deploying a network of radio base stations across a region. The connections between stations form a graph that can be drawn on a plane so that the links intersect only at their endpoints. It is known that any such graph can be colored with four colors so that the endpoints of every link receive distinct colors. Before implementing a four-color algorithm, the engineers decide to solve a simpler problem first: assign one of six available frequencies to each station so that no two stations connected by a link share the same frequency.

*Note: Ignore practical validity—from a Telecommunications Engineering perspective, the problem statement is purely illustrative (a metaphor).*

### Input

The first line of the input contains two integers  $n$  and  $m$ .

The following  $m$  lines contain two integers  $v$  and  $u$  each, stating that there is a bidirectional link between stations  $v$  and  $u$ .

### Output

Output  $n$  integers such that the  $i$ -th integer is between 1 and 6 and describes the frequency assigned to the  $i$ -th station.

### Constraints

$1 \leq n, m \leq 2 \cdot 10^5$

Each link is unique.

There is an undirected path between all pairs of stations.

The graph can be drawn on a plane without the links intersecting.

### Example 1:

| INPUT                                         | OUTPUT    |
|-----------------------------------------------|-----------|
| 5 6<br>1 2<br>2 3<br>3 1<br>2 4<br>2 5<br>4 5 | 2 3 1 2 1 |

### Explanation:

Station 1 receives frequency 2, station 2 receives 3, station 3 receives 1, station 4 receives 2 and station 5 receives 1. No two stations that share a link have the same frequency.

Example 2:

| INPUT                    | OUTPUT  |
|--------------------------|---------|
| 4 3<br>1 2<br>2 3<br>3 4 | 2 1 2 1 |

Explanation:

The network is a path of four stations. Alternating frequencies 2 and 1 is a valid assignment using only two of the six available frequencies.